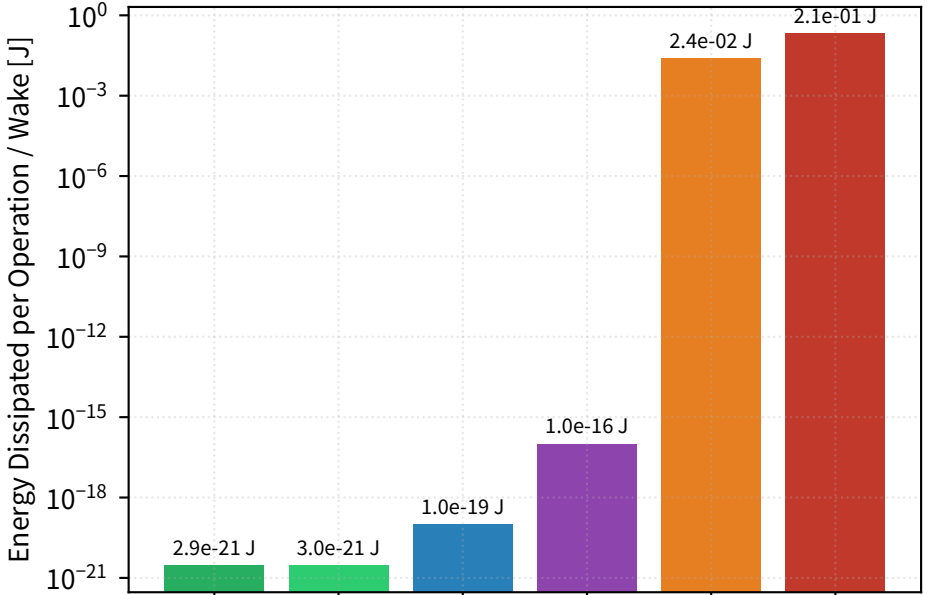
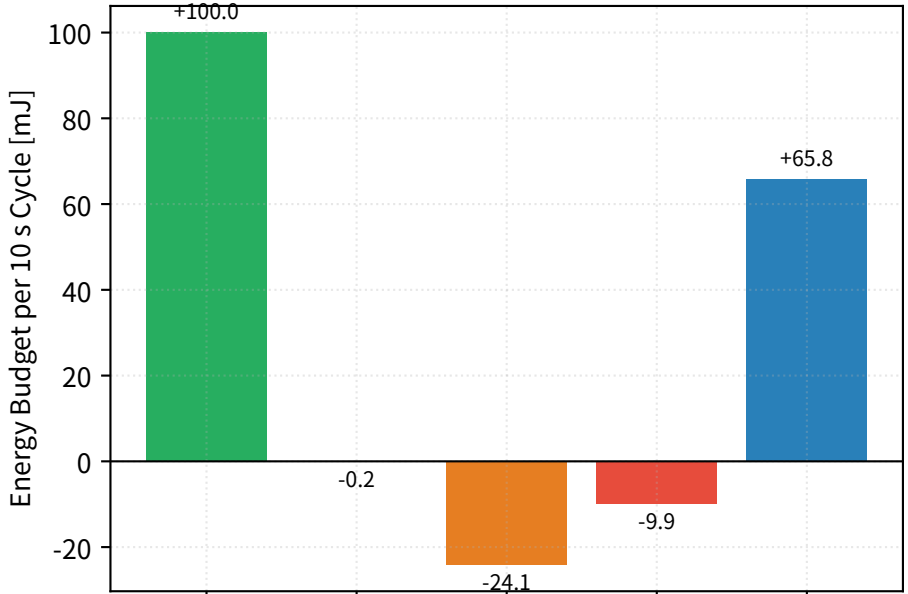

(a) Theoretical Limits vs. Physical Microcontrollers
Energy Dissipated per Operation / Wake [J]
10^{0}
10^{-3}
10^{-6}
10^{-9}
10^{-12}
10^{-15}
10^{-18}
10^{-21}
2.9e-21 J
3.0e-21 J
1.0e-19 J
1.0e-16 J
2.4e-02 J
2.1e-01 J
Landauer Limit
Colloidal SiO2 (Nature 2012)
Superconducting Logic Gate
Advanced CMOS (Logic Gate)
ESP32-C3 v2 (+365ms Wake)
ESP32-C3 v1 (+2s Wake)


(b) Configuration D Energy Balance ($\Delta T = 15\,^{\circ}$C)
Energy Budget per 10 s Cycle [mJ]
100
80
60
40
20
0
-20
+100.0
-0.2
-24.1
-9.9
+65.8
TEG Harvest (10 s)
Deep Sleep (10 s)
Wake Control (v2 ~365 ms)
SOS Optical (100 ms)
Net Surplus (Self-Powered)